

MATH 3220-002 Handout

Summary of Differential Forms

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1 Introduction

Here we give a summary of the differential form formalism. Recall that there are three perspectives on differential forms:

Synthetic: Differential forms are certain strings of symbols subject to certain operational rules.

Differential Calculus: Differential forms are alternating multilinear function valued functions.

Integral Calculus: Differential forms are objects whose signed integrals over chains are well defined.

There are no proofs in this document.

Standing Exercise 1 Find and correct all typos in this document.

Standing Exercise 2 Given any statement in this document, classify the statement as definition, hypothesis, or consequence. Prove all consequences.

Follow-up Exercise Throughout we focus on *real valued* differential forms. Develop the theory of differential forms taking values in an arbitrary normed space, assuming it is finite dimensional or it is endowed with an **appropriate multiplication** when necessary.

2 Synthetic Perspective

For the synthetic perspective we restrict our attention to differential forms defined on open subsets of spaces of tuples.

Let $M \subseteq \mathbb{R}^n$ be an open subset. A **differential form** on M is a finite linear combination of real valued function multiples of finitely many of

$$1, \quad dx_1, \quad dx_2, \quad \dots, \quad dx_m$$

wedged together. Addition and scalar multiplication work as expected. The wedge operation is compatible with these vector space operations, but it also satisfies additional rules, described below.

2.1 Collections of Differential Forms

The collection of all differential forms on M is denoted by

$$\text{Form}(M).$$

For $k \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$, if only C^k real valued functions are allowed in each summand, the resulting subcollection is denoted by

$$C^k\text{-Form}(M).$$

If only exactly p distinct dx_i 's are allowed in each summand, the resulting subcollection is denoted by

$$\text{Form}^p(M).$$

Finally we put

$$C^k\text{-Form}^p(M) = C^k\text{-Form}(M) \cap \text{Form}^p(M).$$

All these collections are vector spaces.

Thus for instance

- Any C^k differential 0-form is of the form

$$\omega = f,$$

where $f \in C^k(M; \mathbb{R})$.

- Any C^k differential 1-form is of the form

$$\omega = f_1 dx_1 + f_2 dx_2 + \cdots + f_n dx_n,$$

where $f_1, f_2, \dots, f_n \in C^k(M; \mathbb{R})$.

- Any C^k differential 2-form is of the form

$$\omega = f_{12} dx_1 \wedge dx_2 + f_{13} dx_1 \wedge dx_3 + \cdots + f_{n-1n} dx_{n-1} \wedge dx_n,$$

where $f_{12}, f_{13}, \dots, f_{n-1n} \in C^k(M; \mathbb{R})$.

- Any C^k differential 3-form is of the form

$$\begin{aligned} \omega = & f_{123} dx_1 \wedge dx_2 \wedge dx_3 \\ & + f_{124} dx_1 \wedge dx_2 \wedge dx_4 \\ & + \cdots \\ & + f_{n-2n-1n} dx_{n-2} \wedge dx_{n-1} \wedge dx_n, \end{aligned}$$

where all coefficient functions are in $C^k(M; \mathbb{R})$.

- Any C^k differential form is of the form

$$\omega = \omega_0 + \omega_1 + \omega_2 + \cdots + \omega_m,$$

where m is a nonnegative integer, and for each $p \in \{0, 1, \dots, m\}$, ω_p is a C^k differential p -form. As ω_i 's are additive components of ω , instead of using the "product" symbol $\prod$, one uses the "direct sum" symbol $\bigoplus$ and writes

$$\begin{aligned} \text{Form}(M) &= \bigoplus_{p=0}^{\infty} \text{Form}^p(M), \\ C^k\text{-Form}(M) &= \bigoplus_{p=0}^{\infty} C^k\text{-Form}^p(M) \end{aligned}$$

to summarize the state of affairs.

2.2 Wedge Product

The **wedge product** on differential forms is a bilinear transformation

$$\wedge : \text{Form}(M) \times \text{Form}(M) \rightarrow \text{Form}(M)$$

subject to the following additional rules:

WedgeRule1: For any $f \in \text{Form}^0(M)$ and for any two $\omega, \eta \in \text{Form}(M)$,

$$f(\omega \wedge \eta) = (f\omega) \wedge \eta = \omega \wedge (f\eta) = f \wedge \omega \wedge \eta = \omega \wedge f \wedge \eta = \omega \wedge \eta \wedge f.$$

WedgeRule2: For any $i \in \{1, 2, \dots, n\}$,

$$dx_i \wedge dx_i = 0.$$

WedgeRule3: For any $i, j \in \{1, 2, \dots, n\}$,

$$dx_i \wedge dx_j = -dx_j \wedge dx_i.$$

(There is a unique bilinear transformation $\wedge$ on differential forms satisfying these rules.)

In particular, if $\omega \in \text{Form}^p(M)$ and $\eta \in \text{Form}^q(M)$, then $\omega \wedge \eta \in \text{Form}^{p+q}(M)$. As wedge products do not affect differentiability, if $\omega \in C^k\text{-Form}^p(M)$ and $\eta \in C^\ell\text{-Form}^q(M)$, then $\omega \wedge \eta \in C^{\min\{k, \ell\}}\text{-Form}^{p+q}(M)$. Note that by **WedgeRule1** above, while one can put a wedge between a function (0-form) and a differential form, as the position of the function does not matter, often this wedge is suppressed. Also note that the third rule implies the second rule.

2.3 Exterior Derivative

The **exterior derivative** on differential forms is a linear transformation

$$d : C^{k+1}\text{-Form}(M) \rightarrow C^k\text{-Form}(M)$$

subject to the following additional rules:

ExtRule1: For any $f \in C^{k+1}\text{-Form}^0(M)$,

$$df = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \cdots + \frac{\partial f}{\partial x_n} dx_n = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i.$$

ExtRule2: For any $f \in C^{k+1}\text{-Form}^0(M)$ and any $i_1, \dots, i_p \in \{1, \dots, n\}$,

$$d(fdx_{i_1} \wedge dx_{i_2} \wedge \cdots \wedge dx_{i_p}) = (df) \wedge dx_{i_1} \wedge dx_{i_2} \wedge \cdots \wedge dx_{i_p}.$$

(There is a unique linear transformation d on differential forms satisfying these rules.)

In particular if $\omega \in C^{k+1}\text{-Form}^p(M)$, then $d\omega \in C^k\text{-Form}^{p+1}(M)$. Two remarks are in order:

- If $k = \infty$, then $k + 1 = \infty$. For this reason, in contexts where regularity is not as important, one considers only C^∞ differential forms (or even beyond that, real analytic or polynomial differential forms).
- There are ways to define the exterior differentiation on C^0 differential forms; ultimately this is done so that a Fundamental Theorem of Calculus and an integration by parts hold. For now (and maybe for this class) we'll not dwell on exterior differentiation of continuous differential forms.

2.3.1 Exact versus Closed Differential forms

The exterior differentiation operator d is a linear transformation, hence its kernel $\ker(d)$ is a vector subspace of $C^1\text{-Form}(M)$ and its image $\text{im}(d)$ is a vector subspace of $\text{Form}(M)$.

- A differential form ω is **exact** if it is in the image of the exterior differentiation operator d , that is,

$$\exists \phi \in C^1\text{-Form}(M) : d\phi = \omega.$$

- A differential form ω is **closed** if it is in the kernel of the exterior differentiation operator d , that is,

$$d\omega = 0.$$

- Two differential forms ω and η are C^k -**cohomologous** if their difference is a C^k exact differential form, that is,

$$\exists \phi \in C^{k+1}\text{-Form}(M) : \omega = \eta + d\phi.$$

2.3.2 Operational Perspective

Operationalizing one step further, one can leverage the wedge product to consider the exterior derivative d *itself* to be a certain differential 1-form with partial differential operator coefficients:

$$d = \frac{\partial}{\partial x_1} dx_1 + \frac{\partial}{\partial x_2} dx_2 + \cdots \frac{\partial}{\partial x_n} dx_n;$$

Then applying the exterior derivative to a differential form would correspond to taking the wedge of exterior derivative with the form:

$$d \wedge \omega = d\omega.$$

This operational perspective makes the verification of $dd = 0$ slightly faster: one computes instead $d \wedge d$ using the wedge product rules (multiplying two operators correspond to the appropriate second order partial derivative).

2.4 Pullback

Let now $N \subseteq \mathbb{R}^m$ be another open subset, $\ell \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$, $\Phi \in C^{\ell+1}(M; N)$. As Φ takes values in a set of tuples, we may write

$$\begin{aligned}\Phi(x_1, x_2, \dots, x_n) = & (\Phi_1(x_1, x_2, \dots, x_n), \\ & \Phi_2(x_1, x_2, \dots, x_n), \\ & \vdots \\ & \Phi_m(x_1, x_2, \dots, x_n)).\end{aligned}$$

Then the **pullback** of differential forms induced by Φ is a linear transformation

$$\Phi^* : C^k\text{-Form}(N) \rightarrow C^{\min\{k, \ell\}}\text{-Form}(M)$$

subject to the following additional rules:

PullbackRule1: For any $f \in \text{Form}^0(M)$,

$$\Phi^*(f) = f \circ \Phi.$$

PullbackRule2: For any $f \in \text{Form}^0(M)$, and any $i_1, \dots, i_p \in \{1, \dots, n\}$

$$\Phi^*(f dx_{i_1} \wedge dx_{i_2} \wedge \dots \wedge dx_{i_p}) = \Phi^*(f) d\Phi_{i_1} \wedge d\Phi_{i_2} \wedge \dots \wedge d\Phi_{i_p}.$$

PullbackRule3: For any $\omega, \eta \in \text{Form}(M)$,

$$\Phi^*(\omega \wedge \eta) = \Phi^*(\omega) \wedge \Phi^*(\eta).$$

PullbackRule4: For any $\omega \in C^{k+1}\text{-Form}(M)$,

$$\Phi^*(d\omega) = d(\Phi^*(\omega)).$$

(There is a unique linear transformation Φ^* on differential forms satisfying these rules.)

Here, the last two rules are in fact implied by the first two rules. Also note that if $\omega \in \text{Form}^p(N)$, then $\Phi^*(\omega) \in \text{Form}^p(M)$. Finally, if $\Psi \in C^{\ell+1}(N; P)$ is yet another function into an open subset $P \subseteq \mathbb{R}^p$, then

$$(\Psi \circ \Phi)^* = \Phi^* \circ \Psi^*.$$

2.5 Hodge Star

The **Hodge star** on differential forms is a linear transformation

$$\star : \text{Form}(M) \rightarrow \text{Form}(M)$$

subject to the following additional rules:

HodgeRule1: For any $\omega \in \text{Form}(M)$ and any $f \in \text{Form}^0(M)$,

$$\star(f\omega) = f \star(\omega).$$

HodgeRule2: For any $i_1, \dots, i_r \in \{1, 2, \dots, n\}$, if these indices are pairwise distinct,

$$dx_{i_1} \wedge \dots \wedge dx_{i_r} \wedge \star(dx_{i_1} \wedge \dots \wedge dx_{i_r}) = dx_1 \wedge dx_2 \wedge \dots \wedge dx_n.$$

(There is a unique linear transformation $\star$ on differential forms satisfying these rules.)

In particular, if $\omega \in \text{Form}^p(M)$ and $p \leq n$, then $\star\omega \in \text{Form}^{n-p}(M)$.

2.5.1 Exterior Coderivative

The **exterior coderivative** on differential forms is the linear transformation

$$d^* = \star d \star : C^{k+1}\text{-Form}(M) \rightarrow C^k\text{-Form}(M).$$

In particular, for any $\omega \in C^{k+1}\text{-Form}^p(M)$, $d^*\omega \in C^k\text{-Form}^{p-1}(M)$. Due to the nature of Hodge star, we have $d^*d^* = 0$ just like the exterior derivative.

2.5.2 Laplace Operator

The **Laplace operator** on differential forms is the linear transformation

$$\Delta = (d + d^*)^2 = (dd^* + d^*d) : C^{k+2}\text{-Form}(M) \rightarrow C^k\text{-Form}(M).$$

(Technically, this is the *analyst's* Laplace operator; the *geometer's* (or *physicist's*) Laplace operator (or the exterior coderivative) has an additional -1 factor. The difference in convention seems to be due to the fact that, down the road, according to our convention the eigenvalues of the Laplace operator end up being nonpositive. Our convention also matches with the convention in Calc III.)

In particular, for any $\omega \in C^{k+2}\text{-Form}^p(M)$, $\Delta\omega \in C^k\text{-Form}^p(M)$. A differential form is **harmonic** if it is in the kernel of the Laplace operator. The **Dirac operator** on differential forms is the linear transformation that is the squareroot of the Laplace operator:

$$\sqrt{\Delta} = (d + d^*) : C^{k+1}\text{-Form}(M) \rightarrow C^k\text{-Form}(M).$$

(To be more accurate, this defines a Dirac operator; in general a Dirac operator is any linear transformation whose square is the

Laplace operator.)

2.5.3 Musical Isomorphisms

Musical isomorphisms are tools to convert differential 1-forms to vector fields and vice versa. The **dièse (or sharp, or index-raising)** operator is the following linear transformation:

$$\sharp : C^k\text{-Form}^1(M) \rightarrow C^k(M; \mathbb{R}^n), \sum_{i=1}^n f_i dx_i \mapsto (f_1, f_2, \dots, f_n).$$

The diese operator is invertible, and its inverse $\flat = \sharp^{-1}$ is called the **bémol (or flat, or index-lowering)** operator. The latter space can be considered to be a space of vector fields on M (in other words the space of first order ordinary differential equations on M).

2.5.4 Calc III Operators

Now suppose $n \in \{1, 2, 3\}$. Let us call a real valued function on M a **scalar field**; thus scalar fields are exactly the same as differential 0-forms.

1D If $n = 1$, we'll use $t = x_1$ as the coordinate, a vector field (or slope field) on M is a function $F \in C^k(M; \mathbb{R})$. The corresponding differential 1-form is $\flat F = F dt$.

- For a scalar field $f \in C^{k+1}(M; \mathbb{R})$, the gradient slope field is

$$\text{grad}(f) = \frac{df}{dt} = \sharp df.$$

- For a slope field $F \in C^{k+1}(M; \mathbb{R})$, the divergence scalar field is

$$\operatorname{div}(F) = \frac{dF}{dt} = \star d \star \flat F.$$

- For a scalar field $f \in C^{k+1}(M; \mathbb{R})$, its Laplacian scalar field is

$$\Delta f = \operatorname{div} \circ \operatorname{grad}(f) = \frac{d^2 f}{dt^2} = \star d \star df = \operatorname{tr}(f'').$$

2D If $n = 2$, we'll use $(x, y) = (x_1, x_2)$ as the coordinates. A vector field on M is a function $F = (P, Q) \in C^k(M; \mathbb{R}^2)$. The corresponding differential 1-form is $\flat(F) = Pdx + Qdy$.

- For a scalar field $f \in C^{k+1}(M; \mathbb{R})$, the gradient vector field is

$$\operatorname{grad}(f) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = \sharp df.$$

- For a vector field $F = (P, Q) \in C^{k+1}(M; \mathbb{R}^2)$, the divergence scalar field is

$$\operatorname{div}(F) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = \star d \star \flat F.$$

- For a scalar field $f \in C^{k+1}(M; \mathbb{R})$, its Laplacian scalar field is

$$\Delta f = \operatorname{div} \circ \operatorname{grad}(f) = \frac{d^2 f}{dx^2} + \frac{d^2 f}{dy^2} = \star d \star df = \operatorname{tr}(f'').$$

3D If $n = 3$, we'll use $(x, y, z) = (x_1, x_2, x_3)$ as the coordinates. A vector field on M is a function $F = (P, Q, R) \in C^k(M; \mathbb{R}^3)$. The corresponding differential 1-form is $\flat F = Pdx + Qdy + Rdz$.

- For a scalar field $f \in C^{k+1}(M; \mathbb{R})$, the gradient vector field is

$$\text{grad}(f) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = \sharp df.$$

- For a vector field $F = (P, Q, R) \in C^{k+1}(M; \mathbb{R}^3)$, the curl vector field is

$$\text{curl}(F) = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) = \sharp \star d\flat F.$$

- For a vector field $F = (P, Q, R) \in C^{k+1}(M; \mathbb{R}^3)$, the divergence scalar field is

$$\text{div}(F) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = \star d \star \flat(F).$$

- For a scalar field $f \in C^{k+1}(M; \mathbb{R})$, its Laplacian scalar field is

$$\Delta f = \text{div} \circ \text{grad}(f) = \frac{d^2 f}{dx^2} + \frac{d^2 f}{dy^2} + \frac{d^2 f}{dz^2} = \star d \star df = \text{tr}(f'').$$

3 Differential Calculus Perspective

We'll now describe differential forms from the perspective of differential calculus as we have developed so far in class. While the differential calculus perspective seems completely independent from the synthetic perspective, these two perspectives are completely and perfectly compatible.

We'll be able to rigorously define everything except Hodge star and the musical isomorphisms in the context of open subsets M of arbitrary normed spaces E . For the integral calculus perspective to differential forms none of these extra structures are needed; we'll use Hodge star only in $\mathbb{R}^n$. Hence we do not discuss the musical isomorphisms, Hodge star and the related operators in detail.

3.1 Collections of Differential Forms

Let $M \subseteq E$ be an open subset of a normed space, $k \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$, $p \in \mathbb{Z}_{\geq 0}$. Then by definition

$$\begin{aligned}\text{Form}^p(M) &= F(M; \mathcal{BA}_p(E; \mathbb{R})), \\ C^k\text{-Form}^p(M) &= C^k(M; \mathcal{BA}_p(E; \mathbb{R})).\end{aligned}$$

Here $\mathcal{BA}_p(E; \mathbb{R})$ is the closed vector subspace of the normed space $\mathcal{BL}_p(E; \mathbb{R})$ of all bounded p -linear transformations $E^p \rightarrow \mathbb{R}$ consisting of only those that are alternating. Thus any $T \in \mathcal{BA}_p(E; \mathbb{R})$ is a bounded p -linear $T : E^p \rightarrow \mathbb{R}$ such that swapping any two input vectors changes the value of T by a factor of -1 . Note also that the word "closed" here is in the sense of metric spaces, so that we really mean: if a sequence of alternating bounded p -linear transformations converge in the multilinear operator norm to some bounded p -linear

transformation, then this limit transformation too will be automatically alternating. By exercises mentioned in class (when we were discussing metric spaces), $\mathcal{BA}_p(E; \mathbb{R})$ is a Banach space (even if E itself is not). We then put as in the syntactic perspective

$$\begin{aligned}\text{Form}(M) &= \bigoplus_{p=0}^{\infty} \text{Form}^p(M), \\ C^k\text{-Form}(M) &= \bigoplus_{p=0}^{\infty} C^k\text{-Form}^p(M);\end{aligned}$$

These notations mean that a differential form on M is again of the form

$$\omega = \omega_0 + \omega_1 + \cdots + \omega_p,$$

where for each i , ω_i is a differential i -form. In other words, we consider the sum of infinitely many differential forms a differential form in and of itself insofar as the degrees of the summands are bounded, and the series is convergent with respect to some reasonable sense of convergence, to be discussed below.

3.1.1 Convergence of Differential Forms

(The contents of this paragraph are more advanced than what we'll need in MATH 3220; I am writing it to give a sense as to what sorts of technicalities are involved down the road.)

The collections of differential forms we defined above are all vector spaces. As is customary in calculus and analysis, we would rather have an additional structure on collections of differential forms so as to be able to at least talk about the convergence or divergence of a sequence of differential forms. There are at least three distinct and

useful ways to accomplish this:

1. On C^k differential p-forms, use C^k uniform metric, and on arbitrary C^k differential forms, take a product metric. The issue with this is that as the subset M is open, we don't have Maximum Principle, so we might have differential forms that are k -times differentiable but *unbounded* derivatives; hence we would need to allow the C^k uniform metric to take ∞ as a value. We had indeed used this metric a few times throughout this class, but not in a deep manner. One exception to this was problem 2 in [problem set 2](#); there we considered functions defined on a compact interval. In single variable calculus derivatives of a function on the boundary are relatively easy to handle; on the other hand in multivariable calculus derivatives on the boundary become a lot more [technical](#); indeed to bypass these technicalities we developed differential calculus for functions between open subsets. Another technicality that I conveniently did not even point at is that in fact even if E were finite dimensional, and even if we restrict our attention to forms with bounded derivatives, spaces of C^∞ forms would [fail to be normed spaces](#).
2. On C^k differential p-forms, use a notion of convergence that has more convergent sequences compared to C^k uniform convergence, but fewer convergent sequences compared to C^k pointwise convergence. Noting that any singleton is compact, we replace "point" with a ["big point"](#) (aka "compact subset"), obtaining " C^k uniform convergence on compacta". More precisely we define C^k **uniform convergence on compacta (or C^k compact-open topology or weak C^k topology)** by declaring a sequence $\omega_\bullet$ of C^k differential p-forms to be **convergent** to ω_∞ if for any compact $K \subseteq M$, the restrictions of $\omega_\bullet$ to K converges to the restriction of ω_∞ in the C^k uniform metric on K . It turns out

again there is no notion of norm that corresponds to this notion of convergence; and at least if E is finite dimensional, there is a metric that corresponds to this notion of convergence.

3. The third option is yet another notion of convergence that has more convergent sequences compared to C^k uniform metric, but fewer compared to the topology of C^k uniform convergence on compacta; the aptly named "strong C^k topology" is defined like so: We define **strong C^k topology (or C^k Whitney topology)** by declaring a sequence $\omega_\bullet$ of C^k differential p -forms to be **convergent** to ω_∞ if it is convergent according to the topology of C^k uniform convergence on compacta, and furthermore

$$\forall \varepsilon \in C^0(M; \mathbb{R}_{>0}), \exists N \in \mathbb{Z}_{\geq 0}, \forall n \in \mathbb{Z}_{\geq N}, \forall x \in M : \\ \sup_{0 \leq i \leq k} \left| \omega_n^{(i)}(x) - \omega_\infty^{(i)}(x) \right|_{\mathcal{BA}_p(E; \mathbb{R})} < \varepsilon(x).$$

One should not expect this last option to be encoded by a metric or norm even in the most reasonable cases.

Each of these options also establish a sense of what it means for a sequence of differential forms to be Cauchy (that is, they define uniformities). All these convergence considerations are too advanced for the purposes of MATH 3220, but still standard issue for a modern mathematician.

3.2 Wedge Product

The **wedge product** on differential forms is a bilinear transformation

$$\wedge : \text{Form}(M) \times \text{Form}(M) \rightarrow \text{Form}(M)$$

defined by the following formula: for any $\omega \in \text{Form}^p(M)$ and any $\eta \in \text{Form}^q(M)$, $\omega \wedge \eta \in \text{Form}^{p+q}(M)$ is defined so that for any $x \in M$, $\omega \wedge \eta(x) : E^{p+q} \rightarrow \mathbb{R}$ is the following alternating bounded p -linear transformation:

$$\begin{aligned} \omega \wedge \eta(x) : (v_1, v_2, \dots, v_p, v_{p+1}, v_{p+2}, \dots, v_{p+q}) \\ \mapsto \frac{1}{p!q!} \sum_{\sigma \in \text{Bij}_{p+q}} \text{sign}(\sigma) \omega(x)(v_{\sigma(1)}, v_{\sigma(2)}, \dots, v_{\sigma(p)}) \\ \eta(x)(v_{\sigma(p+1)}, v_{\sigma(p+2)}, \dots, v_{\sigma(p+q)}), \end{aligned}$$

where Bij_{p+q} is the collection of all permutations of $\{1, 2, \dots, p + q\}$. For each permutation σ , we permute the $p + q$ input vectors, plug the first p input vectors into $\omega(x)$ to obtain a number, plug the remaining q input vectors into $\eta(x)$ to obtain another number, multiply these two numbers, use an appropriate ± 1 factor, add all such numbers over all σ and finally scale appropriately to correct for "overcounted" permutations.

I find this formula rather nasty (note that the complexity of it has nothing to do with E being an arbitrary vector space, but rather the size of p and q). Still, it is not significantly nastier than the formula for **determinants**. The complexity of this formula (as well as that for exterior derivative) is one of the main reasons why we established the synthetic approach before delving into the differential calculus approach to differential forms, of course with the caveat that the synthetic approach requires that one considers only finite dimensions and uses coordinates (which in general may not be available, or desirable).

Let's also mention in passing that if restricted to C^k forms, as the wedge product involve algebraic (hence C^∞) operations, it will again

produce a C^k form, and furthermore with respect to *any* reasonable notion of convergence on forms it will be bounded bilinear.

3.2.1 Wedge Product in Practice

How I like to take wedge products in practice is like so: Let ω be a differential p -form and η be a differential q -form. Fix a position $x \in M$. Then $\omega \wedge \eta(x)$ will be an alternating bounded p -linear function

$$E^{p+q} \rightarrow \mathbb{R}.$$

Let $(v_1, v_2, \dots, v_p, v_{p+1}, v_{p+2}, \dots, v_{p+q}) \in E^{p+q}$ be arbitrary. We need to determine the number that is

$$\omega \wedge \eta(x)(v_1, v_2, \dots, v_p, v_{p+1}, v_{p+2}, \dots, v_{p+q}).$$

I know right from the get go that this number should depend multilinearly on the input vectors. We also have that $\omega(x)$ and $\eta(x)$ themselves depend multilinearly on the input vectors. So we have that

$$\omega(x)(v_1, v_2, \dots, v_p) \eta(x)(v_{p+1}, v_{p+2}, \dots, v_{p+q})$$

is definitely $(p + q)$ -linear in the $p + q$ input vectors. However this expression fails to be *completely* alternating: for instance, switching v_1 and v_2 contributes a factor of -1 due to the alternating nature of $\omega(x)$ itself, but switching v_1 and v_{p+1} will produce a number that in principle has nothing to do with the original number. To remedy this problem, I will add or subtract to this number by permuting the input vectors by hand to force an expression that is alternating (admittedly this part requires some trial and error and hence experience). But then some of the permutations are going to be redundant, as the $\omega(x)$ and $\eta(x)$ parts are already alternating. To account for the redundancy I will divide by the appropriate numbers.

This method for instance establishes the following special cases of the master formula for wedge products:

- If ω_1 and ω_2 are two differential 1-forms on M , then for any $x \in M$ and for any $v_1, v_2 \in E$,

$$\begin{aligned}\omega_1 \wedge \omega_2(x)(v_1, v_2) &= \text{constant} [\omega_1(x)(v_1)\omega_2(x)(v_2) \pm \text{junk to force alternation}] \\ &= \omega_1(x)(v_1)\omega_2(x)(v_2) - \omega_1(x)(v_2)\omega_2(x)(v_1) \\ &= \det \begin{pmatrix} \omega_1(v_1) & \omega_1(v_2) \\ \omega_2(v_1) & \omega_2(v_2) \end{pmatrix}.\end{aligned}$$

- If $\omega_1, \omega_2, \omega_3$ are three differential 1-forms on M , then for any $x \in M$ and for any $v_1, v_2, v_3 \in E$,

$$\begin{aligned}
& \omega_1 \wedge \omega_2 \wedge \omega_3(x)(v_1, v_2, v_3) \\
&= \text{constant} \left[\omega_1(x)(v_1)\omega_2(x)(v_2)\omega_3(x)(v_3) \pm \text{junk to force alternation} \right] \\
&= \text{constant} \left[\omega_1(x)(v_1)\omega_2(x)(v_2)\omega_3(x)(v_3) \right. \\
&\quad \pm \omega_1(x)(v_2)\omega_2(x)(v_1)\omega_3(x)(v_3) \\
&\quad \pm \omega_1(x)(v_3)\omega_2(x)(v_2)\omega_3(x)(v_1) \\
&\quad \pm \omega_1(x)(v_1)\omega_2(x)(v_3)\omega_3(x)(v_2) \\
&\quad \pm \omega_1(x)(v_2)\omega_2(x)(v_3)\omega_3(x)(v_1) \\
&\quad \left. \pm \omega_1(x)(v_3)\omega_2(x)(v_1)\omega_3(x)(v_2) \right] \\
&= \omega_1(x)(v_1)\omega_2(x)(v_2)\omega_3(x)(v_3) \\
&\quad - \omega_1(x)(v_2)\omega_2(x)(v_1)\omega_3(x)(v_3) \\
&\quad - \omega_1(x)(v_3)\omega_2(x)(v_2)\omega_3(x)(v_1) \\
&\quad - \omega_1(x)(v_1)\omega_2(x)(v_3)\omega_3(x)(v_2) \\
&\quad + \omega_1(x)(v_2)\omega_2(x)(v_3)\omega_3(x)(v_1) \\
&\quad + \omega_1(x)(v_3)\omega_2(x)(v_1)\omega_3(x)(v_2) \\
&= \det \begin{pmatrix} \omega_1(v_1) & \omega_1(v_2) & \omega_1(v_3) \\ \omega_2(v_1) & \omega_2(v_2) & \omega_2(v_3) \\ \omega_3(v_1) & \omega_3(v_2) & \omega_3(v_3) \end{pmatrix}.
\end{aligned}$$

(Here in the third equality I chose the constant to be 1 as there were no redundancies.)

- If $\omega_1, \dots, \omega_r$ are r differential 1-forms on M , then for any $x \in M$ and for any $v_1, \dots, v_r \in E$,

$$\omega_1 \wedge \cdots \wedge \omega_r(x)(v_1, \dots, v_r) = \det \begin{pmatrix} \omega_1(v_1) & \omega_1(v_2) & \cdots & \omega_1(v_r) \\ \omega_2(v_1) & \omega_2(v_2) & \cdots & \omega_2(v_r) \\ \vdots & \vdots & \ddots & \vdots \\ \omega_r(v_1) & \omega_r(v_2) & \cdots & \omega_r(v_r) \end{pmatrix}.$$

- If $E = \mathbb{R}^n$ and $1 \leq i_1 < i_2 < \cdots < i_r \leq n$, then for any $x \in M$ and for any $v_1, \dots, v_r \in E$,

$$dx_{i_1} \wedge \cdots \wedge dx_{i_r}(x)(v_1, \dots, v_r) = \det \begin{pmatrix} v_{1i_1} & v_{2i_1} & \cdots & v_{ri_1} \\ v_{1i_2} & v_{2i_2} & \cdots & v_{ri_2} \\ \vdots & \vdots & \ddots & \vdots \\ v_{1i_r} & v_{2i_r} & \cdots & v_{ri_r} \end{pmatrix},$$

where v_{jk} is the k -th entry of v_j .

3.3 Exterior Derivative

The **exterior derivative** on C^{k+1} differential forms on M is a linear transformation

$$d : C^{k+1}\text{-Form}(M) \rightarrow C^k\text{-Form}(M)$$

defined by the following formula: for any $\omega \in C^{k+1}\text{-Form}^p(M)$, $d\omega \in C^k\text{-Form}^{p+1}(M)$ is defined so that for any $x \in M$, $d\omega(x) : E^{p+1} \rightarrow \mathbb{R}$ is the following alternating bounded $(p+1)$ -linear transformation:

$$\begin{aligned}
d\omega(x) &: (v_1, v_2, \dots, v_{p+1}) \\
&\mapsto \sum_{i=1}^{p+1} (-1)^{i+1} \omega'(x)(v_i)(v_1, \dots, v_{i-1}, \widehat{v_i}, v_{i+1}, \dots, v_{p+1}) \\
&= \omega'(x)(v_1)(v_2, v_3, v_4, \dots, v_p, v_{p+1}) \\
&\quad - \omega'(x)(v_2)(v_1, v_3, v_4, \dots, v_p, v_{p+1}) \\
&\quad + \omega'(x)(v_3)(v_1, v_2, v_4, \dots, v_p, v_{p+1}) \\
&\quad \pm \dots \\
&\quad + \pm \omega'(x)(v_{p+1})(v_1, v_2, v_3, \dots, v_{p-1}, v_p).
\end{aligned}$$

Here in the first expression, the $\widehat{v_i}$ paradoxically represents *not writing* that specific vector. Much like the wedge product, exterior derivative will be bounded linear with respect to any reasonable notion of convergence of forms; indeed one can argue that guaranteeing that exterior derivative be bounded linear *ought* to be part of what "reasonable" means in this context.

3.3.1 Exterior Derivative in Practice

I personally find the formula for exterior derivative fairly reasonable, at least when the summation is expanded. One way to remember the formula again is to first come up with one summand that is syntactic and then add additional terms to force alternation. Here is what I mean by this: Let ω be a C^{k+1} differential p -form on M , and fix a position $x \in M$. Then $d\omega(x) : E^{p+1} \rightarrow \mathbb{R}$ will be an alternating $(p+1)$ -linear function. Let $(v_1, \dots, v_{p+1}) \in E^{p+1}$ be arbitrary. We need to determine the number that is

$$d\omega(x)(v_1, \dots, v_{p+1}).$$

While the exterior derivative $d\omega(x)$ at x is not quite the derivative $D\omega(x) = \omega'(x)$, these two notions ought to be related. Note that the latter is, by way of differential calculus, a bounded linear transformation

$$\omega'(x) : E \rightarrow \mathcal{BA}_p(E; \mathbb{R}).$$

Thus if I feed into $\omega'(x)$ a vector, then to the outcome I'll be able to feed p more vectors; the end result will thus be bounded $(p + 1)$ -linear. So I consider the number

$$\omega'(x)(v_1)(v_2, \dots, v_{p+1}).$$

While this is bounded $(p + 1)$ -linear, it fails to be alternating in general. Much like the practical approach to wedge products, I'll then add to this more terms and force alternation. This time it's relatively easier, as the fact that I *chose* the first vector among the tuple of input vectors hints at how I should come up with extra terms. In fact I'll run through each vector v_i in the tuple of input vectors, feed it to $\omega'(x)$ to obtain a bounded p -linear, feed the remaining p vectors in the tuple of input vectors to *this* p -linear, and alternate factors ± 1 when tallying all such expressions.

Again as special cases we have for instance:

- If f is a C^{k+1} differential 0-form, then for any $x \in M$ and any $v \in E$,

$$df(x)(v) = f'(x)(v),$$

so that for C^{k+1} differential 0-forms, exterior derivative *is* the derivative in the sense of differential calculus.

- If ω is a C^{k+1} differential 1-form, then for any $x \in M$ and any $v_1, v_2 \in E$,

$$d\omega(x)(v_1, v_2) = \omega'(x)(v_1)(v_2) - \omega'(x)(v_2)(v_1).$$

- If ω is a C^{k+1} differential 2-form, then for any $x \in M$ and any $v_1, v_2, v_3 \in E$,

$$\begin{aligned} d\omega(x)(v_1, v_2, v_3) &= \omega'(x)(v_1)(v_2, v_3) \\ &\quad - \omega'(x)(v_2)(v_1, v_3) \\ &\quad + \omega'(x)(v_3)(v_1, v_2). \end{aligned}$$

3.3.2 Exact versus Closed Differential forms

With this definition, again we have the **important formula** for (at least) C^2 forms:

$$d^2 = 0.$$

Verification of this using the synthetic approach already showed up in a problem in **problem set 10**; I highly recommend that you verify this from the differential calculus perspective also.

As in the synthetic approach, we define a form to be **exact** if it is the exterior derivative of some form and **closed** if its exterior derivative is zero. Again we have that any C^1 exact differential form is automatically closed.

Warning The exterior derivative is closely related to the derivative in the sense of differential calculus, and consequently applying exterior derivative twice is closely related to the second derivative in the sense of differential calculus. The exterior derivative of a differential 1-form then can be considered as taking the second derivative, without

necessarily taking the first derivative. Indeed, as for a differential 1-form ω on M , for $x \in M$, and for $v_1, v_2 \in E$ we have by definition

$$d\omega(x)(v_1, v_2) = \omega'(x)(v_1)(v_2) - \omega'(x)(v_2)(v_1),$$

$d\omega = 0$ iff the first derivative of ω at x as a bilinear operator is symmetric. The formalism of differential forms is so that the outcome of an exterior derivative is *always alternating*, whereas the outcome of the second derivative in the sense of differential calculus is *always symmetric, and it need not be zero*. We did not prove this latter statement in class, although we did mention some aspects of the proof, and the last problem in **midterm 2** was about this.

3.4 Pullback

Let $N \subseteq F$ be another open subset in another normed space F , $\ell \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$, $\Phi \in C^{\ell+1}(M; N)$. Then the **pullback** of differential forms induced by Φ is a linear transformation

$$\Phi^* : C^k\text{-Form}(N) \rightarrow C^{\min\{k, \ell\}}\text{-Form}(M)$$

defined by the following formula: for any $\omega \in C^k\text{-Form}^p(N)$, $\Phi^*(\omega) \in C^{\min\{k, \ell\}}\text{-Form}^p(M)$ is defined so that for any $x \in M$, $\Phi^*(\omega)(x) : E^p \rightarrow \mathbb{R}$ is the following alternating bounded p -linear transformation:

$$\Phi^*(\omega)(x) : (v_1, \dots, v_p) \mapsto \omega(\Phi(x))(\Phi'(x)v_1, \dots, \Phi'(x)v_p).$$

3.5 Musical Isomorphisms and Hodge Star

We are skipping the differential calculus definitions of the musical isomorphisms, the Hodge star operator and the operators related to

the Hodge star. It suffices here to say that for these more special operators we'll need additional structure on top of a normed space.

- For the musical isomorphisms, we need that E is a Hilbert space, or more generally that there is a symmetric bounded bilinear $\mathbb{B} : E \times E \rightarrow \mathbb{R}$ such that $\flat_{\mathbb{B}} : E \rightarrow \mathcal{BL}(E; \mathbb{R}), v \mapsto \mathbb{B}(v, \bullet)$ is an isomorphism with bounded linear inverse $\flat_{\mathbb{B}}^{-1} = \sharp_{\mathbb{B}}$. This latter is more general as it allows for **Minkowski products** also, to say the least.
- For the Hodge star operator, we need that E is a normed space with additional structure as in the previous item, and furthermore it is finite dimensional and a $\Omega \in C^0\text{-Form}^{\dim(E)}(M)$ is chosen such that for any $x \in M : \Omega(x) \neq 0$.

Once a Hodge star operator is defined, the definitions of the exterior coderivative, Laplace operator and Dirac operator follow exactly as in the synthetic perspective.

4 Integral Calculus Perspective

Finally we describe differential forms from the perspective of integral calculus. This perspective is fully compatible with the synthetic and differential calculus perspectives, as well as the theory of Riemann integration. I'll focus more on the differential calculus perspective, as it makes the discussion clearer.

Let $M \subseteq E$ be an open subset of a normed space, $k \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$, $p \in \mathbb{Z}_{\geq 0}$. A **current** (or **generalized differential form**) on C^k differential forms is a linear transformation

$$T : C^k\text{-Form}(M) \rightarrow \mathbb{R}, \quad \omega \mapsto T(\omega) = \langle T | \omega \rangle$$

that is continuous with respect to some reasonable sense of convergence of C^k differential forms (the most standard choice is the C^k uniform convergence on compacta mentioned above). A **p-current** on C^k differential forms is a current T such that for any $q \in \mathbb{Z}_{\geq 0}$, if $q \neq p$, for any $\omega \in C^k\text{-Form}^q(M)$, $\langle T | \omega \rangle = 0$.

The theory of currents is deep, and as for its foundations, convergence of differential forms is an important matter, we'll not study the spaces of currents in depth. Instead we'll only consider currents that ultimately can be reduced to the Riemann integral on an aligned box after Hodging an appropriate pullback. In other words, the types of currents we'll be interested in are linear combinations of currents that are completely determined by an aligned box and a sufficiently regular function that induces a pullback on differential forms. The former kind will be called a chain and the latter a cell. Ultimately the Riemann integrals we'll use ought to be signed also, and for this we'll need to discuss the notion of orientation before going far.

4.1 On the Bra-Ket Notation

Instead of writing $T(\omega)$ we'll borrow the **bra-ket notation** from physics and write

$$T(\omega) = \langle T | \omega \rangle.$$

Typically in mathematics literature for any such natural bilinear pairings it's common to use $\langle \bullet, \bullet \rangle$, but since throughout the class we used this symbol to denote an inner product I decided to use a slightly different, but similar, symbol. Inner products allow one to combine two vectors from one vector space into a number. Combining a current with a differential form too is so natural that the notation I am using is evocative of an inner product. In this regard, there is no difference between writing T and writing $\langle T | \bullet \rangle$; the implication is that a current is completely determined by the way it transforms differential forms. On the other hand note that there is a subtle psychological difference between writing $\langle \bullet | \omega \rangle$ and ω . Had we developed currents from the ground up instead of differential forms (which is possible, we discuss this a little bit more below), the situation would be the exact opposite. Of course, after one establishes the general theory it becomes irrelevant which of the two comes first.

4.2 Orientation

Let V be a finite dimensional normed space and put $d = \dim(V)$. Suppose $d \geq 1$ and consider the product normed space $V^d = V \times V \times \cdots \times V$. Thus any d -tuple of vectors of V is an element of V^d ; in particular any basis of V is an element of V^d .

The subset $\text{Bases}(V) \subseteq V^d$ of all bases of V is an open subset of V^d , say as a consequence of problem 4 in **problem set 5**. Furthermore, $\text{Bases}(V)$ has exactly two path-connected components. Any one of

these two path-connected components is called an **orientation** on V . V is called **oriented** if one of the two orientations is specified.

4.2.1 Orientations in terms of Invertible Matrices

A practical question is the determination of whether or not two given bases of V are in the same orientation. Suppose $\beta, \gamma \in \text{Bases}(V)$. Then we can consider the matrix representation $\mathcal{M}_{\gamma \leftarrow \beta}(I_V)$ of the identity transformation; in 2270 language this is nothing but the change-of-basis matrix that transform the β -representations of vectors in V into their γ -representations. Then β and γ are in the same orientation iff $\det(\mathcal{M}_{\gamma \leftarrow \beta}(I_V)) > 0$.

4.2.2 Right Hand Rule versus Left Hand Rule

If $V = \mathbb{R}^n$, adhering to the **right hand rule** (or the **counter clockwise direction**) in calculations means that the specified orientation is the path-connected component containing the standard basis; we denote this orientation by $\oplus$. Similarly the **left hand rule** (or the **clockwise direction**) is the other path-connected component; we denote this orientation by $\ominus$.

Let us define the **sign function** for orientations; this will make the integral formulas slightly more convenient to write later:

$$\text{sign} : \{\oplus, \ominus\} \rightarrow \{\pm 1\}$$

$$\oplus \mapsto \begin{cases} +1 & , \text{ if } \oplus = \oplus \\ -1 & , \text{ if } \oplus = \ominus \end{cases}.$$

4.2.3 Orientation-Preserving Diffeomorphisms

Let $(V, \textcircled{?}_V)$ and $(W, \textcircled{?}_W)$ be two oriented finite dimensional inner product spaces, $\tilde{A} \subseteq V$ and $\tilde{B} \subseteq W$ be two open subsets, and $\Phi : \tilde{A} \rightarrow \tilde{B}$ be a C^1 diffeomorphism. Then for any path-connected subset $A \subseteq \tilde{A}$, as $\det(\Phi')$ is continuous and Φ is a diffeomorphism, by the Intermediate Value Theorem, $\{\det(\Phi'(x)) | x \in A\}$ is either a subset of $\mathbb{R}_{>0}$ or a subset of $\mathbb{R}_{<0}$. In the first case we call Φ **orientation-preserving** on A and in the second case we call Φ **orientation-reversing** on A .

Here a word regarding the real valued function $\det(\Phi')$ is in order. Recall that while discussing unsigned Riemann integrals we had made sense of the function $|\det(\Phi')|$ on finite dimensional inner product spaces. Having the additional structure of orientations on V and W now allows us to make sense of the same expression *without* the absolute value. Let $\beta \in \textcircled{?}_V$ and $\gamma \in \textcircled{?}_W$ be two oriented orthonormal bases of V and W , respectively. Then for each $x \in \tilde{A}$, we can compute $\det(\mathcal{M}_{\gamma \leftarrow \beta}(\Phi'(x)))$. Now if $\bar{\beta} \in \textcircled{?}_V$ and $\bar{\gamma} \in \textcircled{?}_W$ is another pair of oriented orthonormal bases, then $\det(\mathcal{M}_{\gamma \leftarrow \beta}(\Phi'(x))) = \det(\mathcal{M}_{\bar{\gamma} \leftarrow \bar{\beta}}(\Phi'(x)))$, so that this number is independent of the choice of oriented orthonormal bases; we define $\det(\Phi'(x))$ to be this number.

4.3 Cells

Let $\ell \in \mathbb{Z}_{\geq 0}$. Then a $C^{\ell+1}$ **p-cell** $C = [\varphi; B, \textcircled{?}]$, where

- $B \subseteq \mathbb{R}^p$ is a standard aligned box,
- $\varphi \in C^{\ell+1}(\tilde{B}; M)$ for some open subset $\tilde{B} \subseteq \mathbb{R}^p$ such that $B \subseteq \tilde{B}$,
- $\textcircled{?} \in \{+, -\}$,

is a p-current on C^k differential p-forms defined by the following formula: for any $\omega \in C^k\text{-Form}^p(M)$:

$$\begin{aligned} \langle C | \omega \rangle &= \text{sign}(\odot) \int_B \star \varphi^*(\omega)(x_1, \dots, x_p) d(x_1, \dots, x_p) \\ &= \begin{cases} \int_B \star \varphi^*(\omega)(x_1, \dots, x_p) d(x_1, \dots, x_p), & \text{if } \odot = \oplus \\ - \int_B \star \varphi^*(\omega)(x_1, \dots, x_p) d(x_1, \dots, x_p), & \text{if } \odot = \ominus \end{cases}, \end{aligned}$$

where the integrals involved are unsigned Riemann integrals as we discussed before in multivariable integral calculus. One way to think of the integrant involved is this: As $\tilde{B}$ is an open subset in $\mathbb{R}^p$, any differential p-form on it is of the form $f dx_1 \wedge dx_2 \wedge \dots \wedge dx_p$ for some real valued function f on $\tilde{B}$. As ω is a differential p-form, by the nature of the pullback, $\varphi^*(\omega)$ too is a differential p-form on $\tilde{B}$, hence there is a unique function $f_{\omega, \gamma} \in C^{\min\{k, \ell\}}(\tilde{B}; \mathbb{R})$ such that $\varphi^*(\omega) = f_{\omega, \gamma} dx_1 \wedge \dots \wedge dx_p$. Then by the nature of the Hodge star operator, $\star \varphi^*(\omega) = f_{\omega, \gamma}$.

4.3.1 Zero-Cells

Opening up the definition for 0-cells, we are required to consider a standard aligned box $B \subseteq \mathbb{R}^0$. $\mathbb{R}^0$ is supposed to be a zero-dimensional vector space, hence it's the vector space that consists only of the zero vector; consequently B too is forced to be the singleton set $\{0\}$. Any function φ defined on a zero dimensional vector space is infinitely differentiable **vacuously**, hence for 0-cells the $C^{\ell+1}$ regularity requirement does not signify much. Further, we may identify $\varphi : \{0\} \rightarrow M$ with the point $\varphi(0) = x \in M$. Thus for 0-cells instead of writing $[\varphi; B, \odot]$ we write simply $\text{sign}(\odot)[x]$. As a 0-cell, $[x]$ is defined like so: for any $f \in \text{Form}^0(M)$,

$$\langle [x] | f \rangle = f(x).$$

In classical terminology, $[x]$ is called the **delta function** at x (or **Dirac delta** at x). One often also sees the following notation, especially in more applied contexts

$$\langle [x] | f \rangle = \int_M f(y) \delta(y - x) dy.$$

Delta functions are not actual functions on M , and one of the major motivations behind the development of the theory of currents was to make sense of them formally.

4.3.2 One-Cells

For 1-cells too we use a slightly more convenient notation. Let $[\varphi; B, \odot]$ be a $C^{\ell+1}$ 1-cell. Thus $B \subseteq \mathbb{R}$ is a compact interval, $\tilde{B} \subseteq \mathbb{R}$ is an open subset containing B , and $\varphi \in C^{\ell+1}(\tilde{B}; M)$. Denote by $\lambda = \min(B)$ and $\rho = \max(B)$. Then we write

$$[\varphi; \rho \leftarrow \lambda] = \text{sign}(\odot)[\varphi; B, \odot] = \begin{cases} [\varphi; B, \odot] & , \text{ if } \odot = \oplus \\ -[\varphi; B, \odot] & , \text{ if } \odot = \ominus \end{cases}.$$

Thus for any C^k differential 1-form ω ,

$$\langle [\varphi; \rho \leftarrow \alpha] | \omega \rangle = \int_{\alpha}^{\rho} \omega(\varphi(t)) \varphi'(t) dt,$$

the latter integral being signed Riemann integral exactly as in MATH 3210.

4.3.3 Parameterization Independence

The notation $C = [\varphi; B, \textcircled{?}]$ is meant to signify the subtle roles the parameterization φ and the oriented box $(B, \textcircled{?})$ play in defining the cell C . Namely, cells are currents that are defined through subsets that are differentiably parameterized by oriented aligned boxes in Euclidean spaces, but changing the parameterization or the aligned box can change the cell only by a factor of ± 1 .

More rigorously, let B be a standard aligned box in $\mathbb{R}^p$, $\tilde{B}$ be an open subset containing B , and $\textcircled{?} \in \{\textcircled{+}, \textcircled{-}\}$, $\tilde{A}$ be another open subset of $\mathbb{R}^p$ (or an arbitrary oriented finite dimensional inner product space), $\Phi \in \text{Diff}^{\ell+1}(\tilde{B}; \tilde{A})$, and $\varphi \in C^{\ell+1}(\tilde{A}; M)$. We then have that $[\varphi \circ \Phi; B, \textcircled{?}]$ is a p -cell in M .

We now would like to relate the unsigned Riemann integral defining the p -cell $C = [\varphi \circ \Phi; B, \textcircled{?}]$ to an unsigned Riemann integral over $\Phi(B)$. As Φ is not necessarily affine, we are not guaranteed that $\Phi(B)$ is an aligned box; still it is still guaranteed to be compact and Jordan contentable (a fact we stated but did not prove in class), hence unsigned Riemann integrals as well as change of variables are available to us to establish the relation we want. Let us put

$$\text{sign}(\Phi(\textcircled{?})) = \begin{cases} \text{sign}(\textcircled{?}) & , \text{ if } \Phi \text{ is orientation-preserving} \\ -\text{sign}(\textcircled{?}) & , \text{ if } \Phi \text{ is orientation-reversing} \end{cases}.$$

Let $\omega \in C^k\text{-Form}^p(M)$. Then we have

$$\begin{aligned}
& f_{\omega, \varphi \circ \Phi} dx_1 \wedge \cdots \wedge dx_p \\
&= (\varphi \circ \Phi)^*(\omega) \\
&= \Phi^*(\varphi^*(\omega)) \\
&= \Phi^*(f_{\omega, \varphi} dy_1 \wedge \cdots \wedge dy_p) \\
&= f_{\omega, \varphi} \circ \Phi d\Phi_1 \wedge \cdots \wedge d\Phi_p \\
&= f_{\omega, \varphi} \circ \Phi \det(\Phi') dx_1 \wedge \cdots \wedge dx_p,
\end{aligned}$$

so that

$$f_{\omega, \varphi \circ \Phi} = f_{\omega, \varphi} \circ \Phi \det(\Phi'),$$

or equivalently, using Hodge star,

$$\star(\varphi \circ \Phi)^*(\omega) = \Phi^*(\star\varphi^*\omega) \det(\Phi').$$

Using this identity together with the change of variables formula for *unsigned* Riemann integrals, we have

$$\begin{aligned}
\langle C | \omega \rangle &= \langle [\varphi \circ \Phi; B, (?)] | \omega \rangle \\
&= \text{sign}(\odot) \int_B \star(\varphi \circ \Phi)^*(\omega)(x_1, \dots, x_p) d(x_1, \dots, x_p) \\
&= \text{sign}(\odot) \int_B \star(\varphi^*\omega)(\Phi(x_1, \dots, x_p)) \det(\Phi'(x_1, \dots, x_p)) d(x_1, \dots, x_p) \\
&= \text{sign}(\Phi(\odot)) \int_B \star(\varphi^*\omega)(\Phi(x_1, \dots, x_p)) |\det(\Phi'(x_1, \dots, x_p))| d(x_1, \dots, x_p) \\
&= \text{sign}(\Phi(\odot)) \int_{\Phi(B)} \star(\varphi^*\omega)(y_1, \dots, y_p) d(y_1, \dots, y_p),
\end{aligned}$$

so that defining $[\varphi; \Phi(B), \Phi(\odot)]$ by

$$\begin{aligned} & \langle [\varphi; \Phi(B), \Phi(\odot)] | \omega \rangle \\ &= \text{sign}(\Phi(\odot)) \int_{\Phi(B)} \star(\varphi^* \omega)(y_1, \dots, y_p) \, d(y_1, \dots, y_p), \end{aligned}$$

we have that

$$C = [\varphi \circ \Phi; B, \odot] = [\varphi; \Phi(B), \Phi(\odot)],$$

in other words, even if for the latter current, the parameterization is not necessarily defined in a neighborhood of a standard aligned box, and instead a general compact Jordan contentable subset, both of these currents act the same way on differential forms, hence they *are* the same cell. The fact that the same cell has multiple presentations like this is summarized by stating that cells are **independent of parameterization**, so that they are truly objects one can associate to the open subset M itself. Our notation and definition for cells are so that this is not a surprise but is to be expected.

4.3.4 Open Subsets Containing Aligned Boxes

The fact that a cell does not depend on the open subset $\tilde{B}$ on which φ is defined too is part of parameter independence; establishing this would require a separate, somewhat tedious, argument (after all, two open subsets that contain an aligned box need not even be homeomorphic, but certainly as one gets closer and closer from without to the aligned box in the intersection, **eventually** the parameterizations would differ by a diffeomorphism.).

4.3.5 Using More or Less General Aligned Boxes

In light of parameterization independence, one may use larger or smaller collections of Jordan contentable subsets in the definition of cells. For instance

- In our textbook p-cells are required to be defined through φ defined on $I^p = [0, 1] \times \cdots \times [0, 1]$ (except for 1-cells).
- Allow B to be a β -aligned box in some oriented p-dimensional inner product space V with basis β .

Above we opted for an option that is in between these two options, to obtain a balance of expressivity and concreteness.

4.3.6 Classical Notation

In classical notation, one often suppresses part of the data that constitute the cell C, and writes e.g.

$$\langle C | \omega \rangle = \pm \int_{\varphi} \omega = \pm \int_{\varphi(B)} \omega,$$

adjusting $\pm$ based on the context.

Warning In Calc III, integrals are used in two senses; for instance one can take "the line integral of a scalar field with respect to arc length", or one can take "the line integral of a vector field". The integrals of the former kind are really unsigned Riemann integrals of Riemann integrable functions as we discussed in integral calculus, whereas the integrals of the latter kind correspond to integrals of differential forms. In practice the way to distinguish the two kinds is by determining how the integral changes under u-substitutions: if the

correction factor is always positive, then it's an unsigned integral; if the correction factor could be negative, then it's a signed integral.

4.3.7 Integrals and Synthetic Approach

Say $E = \mathbb{R}^n$. Then we can use the synthetic approach to come up with a more concrete formula. Let $\omega \in C^k\text{-Form}^p(M)$. Then ω is of the form

$$\omega = \sum_{1 \leq i_1 < i_2 < \dots < i_p \leq n} f_{i_1 i_2 \dots i_p} dx_{i_1} \wedge dx_{i_2} \wedge \dots \wedge dx_{i_p}.$$

Next, we can consider $\varphi : \tilde{B} \rightarrow M$ componentwise:

$$\begin{aligned} \varphi(x_1, x_2, \dots, x_p) = & (\varphi_1(x_1, x_2, \dots, x_p), \\ & \varphi_2(x_1, x_2, \dots, x_p), \\ & \vdots \\ & \varphi_n(x_1, x_2, \dots, x_p)). \end{aligned}$$

Thus applying the pullback induced by φ to ω synthetically we obtain:

$$\begin{aligned}
& \varphi^*(\omega) \\
&= \varphi^* \left(\sum_{1 \leq i_1 < i_2 < \dots < i_p \leq n} f_{i_1 i_2 \dots i_p} dx_{i_1} \wedge dx_{i_2} \wedge \dots \wedge dx_{i_p} \right) \\
&= \sum_{1 \leq i_1 < i_2 < \dots < i_p \leq n} \varphi^* \left(f_{i_1 i_2 \dots i_p} dx_{i_1} \wedge dx_{i_2} \wedge \dots \wedge dx_{i_p} \right) \\
&= \sum_{1 \leq i_1 < i_2 < \dots < i_p \leq n} \varphi^*(f_{i_1 i_2 \dots i_p}) d\varphi_{i_1} \wedge d\varphi_{i_2} \wedge \dots \wedge d\varphi_{i_p} \\
&= \sum_{1 \leq i_1 < i_2 < \dots < i_p \leq n} \varphi^*(f_{i_1 i_2 \dots i_p}) \left(\sum_{j_1=1}^p \frac{\partial \varphi_{i_1}}{\partial x_{j_1}} dx_{j_1} \right) \wedge d\varphi_{i_2} \wedge \dots \wedge d\varphi_{i_p} \\
&= \dots \\
&= \sum_{1 \leq i_1 < i_2 < \dots < i_p \leq n} \varphi^*(f_{i_1 i_2 \dots i_p}) \\
&\quad \sum_{j_1=1}^p \dots \sum_{j_p=1}^p \frac{\partial \varphi_{i_1}}{\partial x_{j_1}} \frac{\partial \varphi_{i_2}}{\partial x_{j_2}} \dots \frac{\partial \varphi_{i_p}}{\partial x_{j_p}} \\
&\quad dx_{j_1} \wedge dx_{j_2} \wedge \dots \wedge dx_{j_p} \\
&= \sum_{1 \leq i_1 < i_2 < \dots < i_p \leq n} \varphi^*(f_{i_1 i_2 \dots i_p}) \det((\varphi_{i_1}, \varphi_{i_2}, \dots, \varphi_{i_p})') \\
&\quad dx_1 \wedge dx_2 \wedge \dots \wedge dx_p,
\end{aligned}$$

where the determinant involved is the determinant (*without* absolute values) of the derivative of the function

$$\begin{aligned}
 (\varphi_{i_1}, \varphi_{i_2}, \dots, \varphi_{i_p}) : \tilde{B} &\rightarrow \mathbb{R}^p, \\
 (x_1, \dots, x_p) &\mapsto (\varphi_{i_1}(x_1, x_2, \dots, x_p), \\
 &\quad \varphi_{i_2}(x_1, x_2, \dots, x_p), \\
 &\quad \vdots \\
 &\quad \varphi_{i_p}(x_1, x_2, \dots, x_p)).
 \end{aligned}$$

Thus in this case the integral would be

$$\begin{aligned}
 \langle C | \omega \rangle &= \pm \sum_{1 \leq i_1 < i_2 < \dots < i_p \leq n} \int_B f_{i_1 i_2 \dots i_p}(\varphi(x_1, \dots, x_p)) \\
 &\quad \det((\varphi_{i_1}, \varphi_{i_2}, \dots, \varphi_{i_p})'(x_1, \dots, x_p)) \\
 &\quad d(x_1, \dots, x_p),
 \end{aligned}$$

with $\pm$ adjusted according to the context.

4.4 Chains

A $C^{\ell+1}$ **p-chain** on M is a finite linear combination of $C^{\ell+1}$ p-cells on M , all considered as p-currents. A $C^{\ell+1}$ **chain** on M is a current of the form

$$T = C_0 + C_1 + C_2 + \dots + C_p,$$

where for each i , C_i is a $C^{\ell+1}$ i-chain. We denote by $C^{\ell+1}\text{-Chain}^p(M)$ the vector space of all currents made up of all $C^{\ell+1}$ p-chains on M .

and by $C^{\ell+1}\text{-Chain}(M)$ the vector space of all currents made up of all $C^{\ell+1}$ chains on M . Similar to differential forms, we write

$$C^{\ell+1}\text{-Chain}(M) = \bigoplus_{p=0}^{\infty} C^{\ell+1}\text{-Chain}^p(M).$$

4.4.1 Synthetic Approach to Chains

There is also a synthetic approach to chains; often developed in classes that focus more on topology. Thus chains are certain formal expressions subject to certain rules, and instead of the exterior derivative d one uses the boundary operator ∂ . One important distinction even from the syntactic perspective is that there is no **operation** that works as well as wedge product does for differential forms. Also, typically instead of considering aligned boxes one considers **simplices**.

For us, chains are first and foremost currents, that is, linear transformations that transform differential forms into real numbers; for this reason we needed our chains and cells to be at least C^1 . On the other hand, from a topological perspective if one is to develop a synthetic approach to chains, there is no need to keep track (or make sense) of differentiability. Hence from this perspective often one considers arbitrary metric spaces for M , and C^0 chains in M defined similarly. A very fruitful endeavor that is a mixture of this topological and the calculus approach is to consider chains that are more than C^0 , e.g. Lipschitz or rectifiable chains as you studied in **problem set 3**.

4.4.2 Some Currents That Are Chains

Here are some examples of currents that are chains:

- Let $x_1, x_2 \in M$, $c_1, c_2 \in \mathbb{R}$. Then $c_1[x_1] + c_2[x_2]$ is not a 0-cell in M , but it is a 0-chain; it acts on differential 0-forms like so:

$$\langle c_1[x_1] + c_2[x_2] | f \rangle = c_1 \langle [x_1] | f \rangle + c_2 \langle [x_2] | f \rangle = c_1 f(x_1) + c_2 f(x_2).$$

- Let $a, b \in \mathbb{R}$, $I \subseteq \mathbb{R}$ be an open interval with $a, b \in I$, $\gamma \in C^0(I; M)$ be $C^{\ell+1}$ except at finitely many points. Then $C = [\gamma; b \leftarrow a]$ is not a $C^{\ell+1}$ 1-cell, but it is a $C^{\ell+1}$ 1-chain that acts on differential 1-forms like so:

$$\langle C | \omega \rangle = \int_a^b \omega(\gamma(t)) \gamma'(t) dt.$$

4.4.3 Some Currents That Are Not Chains

Many objects we discussed throughout this class (as well as in MATH 3210) can be considered as currents. Here are some examples of currents that are more complicated than chains.

- Let $\gamma \in C^0(\mathbb{R}; M)$. Then $\langle C | f \rangle = \int_0^1 \langle [\gamma(t)] | f \rangle dt = \int_0^1 f \circ \gamma(t) dt$ defines a 0-current C that is not a 0-chain (nor a 1-chain). Thus signed (or unsigned) Riemann integrals along paths are 0-currents.
- Let $x_\bullet : \mathbb{Z}_{\geq 0} \rightarrow M$ be a convergent sequence. Then $\langle C | f \rangle = \sum_{n=0}^{\infty} \langle [x_n] | f \rangle e^{-n} = \sum_{n=0}^{\infty} f(x_n) e^{-n}$ defines a 0-current C that is not a 0-chain.
- Let $C \in C^0([0, 1]; C^1\text{-Chain}^1(M))$. Then $\langle \hat{C} | \omega \rangle = \int_0^1 \langle [C(t)] | \omega \rangle dt$ defines a 1-current $\hat{C}$ that is not a 1-chain.
- Let C be a $(p + q)$ -chain and η be a differential q -form. Then $\langle (C, \eta) | \omega \rangle = \langle C | \omega \wedge \eta \rangle$ defines a p -current (C, η) that is not a p -chain.

4.5 Boundary Operator

The **boundary operator** on $C^{\ell+1}$ chains on M is a linear transformation

$$\partial : C^{\ell+1}\text{-Chain}(M) \rightarrow C^{\ell+1}\text{-Chain}(M)$$

defined as follows: the boundary of any 0-chain is by definition the zero current. Let $C = [\varphi; B, \textcircled{?}]$ be a $C^{\ell+1}$ p -cell with $p \geq 1$. Then by definition

$$B = I_1 \times I_2 \times \cdots \times I_p = [\lambda_1, \rho_1] \times [\lambda_2, \rho_2] \times \cdots \times [\lambda_p, \rho_p],$$

where for each i , $-\infty \leq \lambda_i \leq \rho_i \leq \infty$. The topological boundary of B is

$$\begin{aligned}
\text{bdry}(B) &= \text{bdry}(I_1) \times I_2 \times \cdots \times I_p \\
&\cup I_1 \times \text{bdry}(I_2) \times \cdots \times I_p \\
&\cup \cdots \\
&\cup I_1 \times I_2 \times \cdots \times \text{bdry}(I_p) \\
&= \{\lambda_1, \rho_1\} \times I_2 \times \cdots \times I_p \\
&\cup I_1 \times \{\lambda_2, \rho_2\} \times \cdots \times I_p \\
&\cup \cdots \\
&\cup I_1 \times I_2 \times \cdots \times \{\lambda_p, \rho_p\} \\
&= \{\rho_1\} \times I_2 \times \cdots \times I_p \\
&\cup \{\lambda_1\} \times I_2 \times \cdots \times I_p \\
&\cup I_1 \times \{\rho_2\} \times \cdots \times I_p \\
&\cup I_1 \times \{\lambda_2\} \times \cdots \times I_p \\
&\cup \cdots \\
&\cup I_1 \times I_2 \times \cdots \times \{\rho_p\} \\
&\cup I_1 \times I_2 \times \cdots \times \{\lambda_p\}.
\end{aligned}$$

Any two of these components either don't intersect, or if they do intersect, their intersection has $(p - 1)$ -dimensional Jordan content zero. The boundary of $C = [\varphi; B, \textcircled{?}]$ finally is defined in terms of these topological boundary components:

$$\begin{aligned}
\partial(C) &= \partial[\varphi; B, \textcircled{?}] \\
&= [\varphi; \{\rho_1\} \times I_2 \times \cdots \times I_p, \textcircled{?}] \\
&\quad - [\varphi; \{\lambda_1\} \times I_2 \times \cdots \times I_p, \textcircled{?}] \\
&\quad - [\varphi; I_1 \times \{\rho_2\} \times \cdots \times I_p, \textcircled{?}] \\
&\quad + [\varphi; I_1 \times \{\lambda_2\} \times \cdots \times I_p, \textcircled{?}] \\
&\quad \dots \\
&\quad - (-1)^p [\varphi; I_1 \times I_2 \times \cdots \times \{\rho_p\}, \textcircled{?}] \\
&\quad + (-1)^p [\varphi; I_1 \times I_2 \times \cdots \times \{\lambda_p\}, \textcircled{?}].
\end{aligned}$$

Here each summand ought to be interpreted as a $C^{\ell+1}$ $(p-1)$ -cell. For instance, the first summand is $[\varphi; \{\rho_1\} \times I_2 \times \cdots \times I_p, \textcircled{?}]$. Here $I_2 \times \cdots \times I_p$ is an aligned box in $\mathbb{R}^{p-1}$, it's contained in the open subset $\{(x_2, \dots, x_p) \in \mathbb{R}^{p-1} \mid (\rho_1, x_2, \dots, x_p) \in \tilde{B}\}$, and the function $(x_2, \dots, x_p) \mapsto \varphi(\rho_1, x_2, \dots, x_p)$ is a $C^{\ell+1}$ function from this open subset to M . Thus for each $\eta \in C^k\text{-Form}^{p-1}(M)$, we have

$$\begin{aligned}
&\langle [\varphi; \{\rho_1\} \times I_2 \times \cdots \times I_p, \textcircled{?}] \mid \eta \rangle \\
&= \text{sign}(\textcircled{?}) \int_{I_2 \times \cdots \times I_p} \star \varphi^*(\eta)(\rho_1, x_2, \dots, x_p) d(x_2, \dots, x_p).
\end{aligned}$$

4.5.1 Boundary Operator in Practice

The way I remember this formula is by drawing and writing down the cases for $p = 1, 2, 3$:

Boundary of a 1-Cell The 1-cell $[\gamma; \beta \leftarrow \alpha]$ is about integration from α to β . The topological boundary is $\{\alpha, \beta\}$. As the integration is from α

to β , the orientation of $[\alpha]$ ought to be inward and the orientation of $[\beta]$ ought to be outward. Hence

$$\partial[\gamma; \beta \leftarrow \alpha] = [\gamma(\beta)] - [\gamma(\alpha)].$$

(Of course, ultimately the reason why I interpret inward as $\ominus$ is because I want to be able to use the boundary operator in the Fundamental Theorem of Calculus.)

Boundary of a 2-Cell Let $C = [\gamma; B, \textcircled{?}]$ be a 2-cell. Thus $B = [\lambda_1, \rho_1] \times [\lambda_2, \rho_2]$. and the topological boundary of B consists of the union of the following:

- Bottom edge $[\lambda_1, \rho_1] \times \{\lambda_2\}$
- Right edge $\{\rho_1\} \times [\lambda_2, \rho_2]$
- Top edge $[\lambda_1, \rho_1] \times \{\rho_2\}$
- Left edge $\{\lambda_1\} \times [\lambda_2, \rho_2]$.

If $\textcircled{?} = \textcircled{+}$, B is oriented in the counter clockwise, and hence these four edges ought to be oriented so as to keep B always on the left when traversing any one of them in the chosen orientation. Thus

$$\begin{aligned}
\partial[\gamma; B, \oplus] &= [\gamma; \text{right}, \oplus] \\
&\quad - [\gamma; \text{left}, \oplus] \\
&\quad - [\gamma; \text{top}, \oplus] \\
&\quad + [\gamma; \text{bottom}, \oplus] \\
&= [\gamma; \{\rho_1\} \times [\lambda_2, \rho_2], \oplus] \\
&\quad - [\gamma; \{\lambda_1\} \times [\lambda_2, \rho_2], \oplus] \\
&\quad - [\gamma; [\lambda_1, \rho_1] \times \{\rho_2\}, \oplus] \\
&\quad + [\gamma; [\lambda_1, \rho_1] \times \{\lambda_2\}, \oplus] \\
&= [\gamma; (\rho_1, \rho_2) \leftarrow (\rho_1, \lambda_2)] \\
&\quad - [\gamma; (\lambda_1, \rho_2) \leftarrow (\lambda_1, \lambda_2)] \\
&\quad - [\gamma; (\rho_1, \rho_2) \leftarrow (\lambda_1, \rho_2)] \\
&\quad + [\gamma; (\rho_1, \lambda_2) \leftarrow (\lambda_1, \lambda_2)] \\
&= [\gamma; (\rho_1, \lambda_2) \leftarrow (\lambda_1, \lambda_2)] \\
&\quad + [\gamma; (\rho_1, \rho_2) \leftarrow (\rho_1, \lambda_2)] \\
&\quad + [\gamma; (\lambda_1, \rho_2) \leftarrow (\rho_1, \rho_2)] \\
&\quad + [\gamma; (\lambda_1, \lambda_2) \leftarrow (\lambda_1, \rho_2)].
\end{aligned}$$

If on the other hand $\textcircled{+} = \textcircled{-}$, then all the sides should be oriented the other way around, so as to have B always on the right when traversing along the topological boundary.

Boundary of a 3-Cell In the case of a 3-cell, a standard aligned box in $\mathbb{R}^3$ is involved. This box has exactly 6-faces. Sketching it in the xyz-coordinates, the topological boundary should be oriented so that

$$\partial [3\text{-cell}] = + [\text{right}] - [\text{left}] - [\text{front}] + [\text{back}] + [\text{top}] - [\text{bottom}] .$$

4.5.2 Cycles versus Boundaries among Chains

The boundary operator is a linear transformation, hence its kernel $\ker(\partial)$ as well as its image $\text{im}(\partial)$ are both vector subspaces of $\text{Chain}(M)$.

- A chain C is a **boundary** if it is in the image of the boundary operator, that is,

$$\exists K \in C^{\ell+1}\text{-Chain}(M) : \partial K = C.$$

- A chain C is a **cycle** if it is in the kernel of the boundary operator, that is,

$$\partial C = 0.$$

- Two chains C, D are $C^{\ell+1}$ **homologous** if their difference is a $C^{\ell+1}$ boundary, that is,

$$\exists K \in C^{\ell+1}\text{-Chain}(M) : C = D + \partial K.$$

In a way analogous to the same equation for the exterior derivative, we have

$$\partial^2 = 0,$$

hence any boundary is automatically a cycle. In accounts that prioritize currents over differential forms, often a differential form is called a **cochain**, an exact differential form a **coboundary** and a closed differential form a **cocycle**.

4.5.3 Periods

Let $\omega \in C^k\text{-Form}^p(M)$. The set $\text{Per}(\omega) = \{\langle C | \omega \rangle \mid C \in \ker(\partial)\}$ is the set of **periods** of ω .

4.6 Pushforward

Let $N \subseteq F$ be another open subset in another normed space F , $m \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$, $\Phi \in C^{m+1}(M; N)$. Then the **pushforward** of $C^{\ell+1}$ chains induced by Φ is a linear transformation

$$\Phi_* : C^{\ell+1}\text{-Chain}(M) \rightarrow C^{\min\{m, \ell\}+1}\text{-Chain}(M)$$

defined by the following formula:

$$\Phi_* : [\varphi; B, \odot] \mapsto [\Phi \circ \varphi; B, \odot].$$

Note that in particular for any $C \in C^{\ell+1}\text{-Chain}^p(M)$, $\Phi_*(C) \in C^{\min\{m, \ell\}+1}\text{-Chain}(M)$. Further we have, for any $C \in C^{\ell+1}\text{-Chain}^p(M)$ and any $\omega \in C^k\text{-Form}^p(M)$,

$$\langle \Phi_*(C) | \omega \rangle = \langle C | \Phi^*(\omega) \rangle.$$

An important property of pushforwards is that they commute with the boundary operator, in full analogy with pullbacks and exterior derivative; thus for any $C \in C^{\ell+1}\text{-Chain}(M)$:

$$\Phi_*\partial(C) = \partial\Phi_*(C).$$

5 Fundamental Theorems of Calculus

In MATH 3210, you learned two fundamental theorems of calculus:

- FTC-1 ($\int d$) is about the integral of the derivative.
- FTC-2 ($d \int$) is about the derivative of the integral.

In multivariable calculus too there are analogs of these results. The first result goes by the name of the **Stokes Theorem**, whereas the second result is known as the **Poincaré Lemma**.

5.1 FTC's in 1D

To recapitulate, here are rigorous statements of the two fundamental theorems of calculus from MATH 3210. These statements can be strengthened using the methods and concepts discussed in MATH 3210, but for our purposes they will be sufficient.

Theorem 1 (FTC-1 ($\int d$)): Let $I \subseteq \mathbb{R}$ be an open interval, $k \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$, $\alpha, \beta \in I$. Then

1. For any $f \in C^k(I; \mathbb{R})$ and for any $\phi \in C^{k+1}(I; \mathbb{R})$, if $\phi' = f$, then $\int_{\alpha}^{\beta} f(t)dt = \phi(\beta) - \phi(\alpha)$.
2. For any $\phi \in C^{k+1}(I; \mathbb{R})$, $\int_{\alpha}^{\beta} \phi'(t)dt = \phi(\beta) - \phi(\alpha)$.

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Theorem 2 (FTC-2 ($d \int$)): Let $I \subseteq \mathbb{R}$ be an open interval, $k \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$. Then for any $f \in C^k(I; \mathbb{R})$, and for any $\gamma \in I$, $\left[F_{\gamma} : t \mapsto \int_{\gamma}^t f(s)ds \right] \in C^{k+1}(I; \mathbb{R})$ and $F'_{\gamma} = f$.

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5.2 Stokes Theorem

Theorem 3: Let $M \subseteq E$ be an open subset in a normed space, $k, \ell \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$, $p \in \mathbb{Z}_{\geq 0}$. Then for any $C^{\ell+1}$ $(p+1)$ -chain C in M and for any C^{k+1} differential p -form ω ,

$$\langle \partial C | \omega \rangle = \langle C | d\omega \rangle.$$

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5.3 Poincaré Lemma

To state Poincaré Lemma we need one more definition. For any $x \in M$, we define the **visibility set** $\text{vis}_x(M)$ of x by

$$\text{vis}_x(M) = \{y \in M \mid \forall t \in [0, 1] : (1-t)x + ty \in M\}.$$

For any $x \in M$, $\text{vis}_x(M)$ is a nonempty open subset of M .

For any $x \in M$, we define the **homotopy operator** based at x to be the linear transformation

$$H_x : C^k\text{-Form}(M) \rightarrow C^k\text{-Form}(\text{vis}_x(M))$$

defined as follows: Let $\omega \in C^k\text{-Form}^p(M)$. If $p = 0$, we define $H_x(\omega) = 0$. If $p \geq 1$, then $H_x(\omega)$ is the differential $(p-1)$ -form on $\text{vis}_x(M)$ defined so that for any $y \in \text{vis}_x(M)$ and for any $v_1, \dots, v_{p-1}$,

$$\begin{aligned} H_x(\omega)(y)(v_1, v_2, \dots, v_{p-1}) \\ = \int_0^1 t^{p-1} \omega((1-t)x + ty)(y-x, v_1, v_2, \dots, v_{p-1}) dt. \end{aligned}$$

A straightforward regularity argument guarantees that $H_x(\omega) \in$

$C^k\text{-Form}^{p-1}(\text{vis}_x(M))$.

Theorem 4: Let $M \subseteq E$ be an open subset in a normed space, $k \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$, $p \in \mathbb{Z}_{\geq 0}$, $x \in M$. Then

1. For any $f \in C^{k+1}\text{-Form}^0(M)$,

$$H_x(df) = f - f(x)$$

as differential 0-forms on $\text{vis}_x(M)$.

2. For any $\omega \in C^{k+1}\text{-Form}^{p+1}(M)$,

$$dH_x(\omega) + H_x(d\omega) = \omega$$

as differential $(p+1)$ -forms on $\text{vis}_x(M)$.

3. For any $\omega \in C^{k+1}\text{-Form}^{p+1}(M)$, if ω is closed as a differential $(p+1)$ -form on $\text{vis}_x(M)$, then

$$dH_x(\omega) = \omega$$

as differential $(p+1)$ -forms on $\text{vis}_x(M)$. Consequently $H_x(\omega) \in C^{k+2}\text{-Form}^p(\text{vis}_x(M))$ and ω is exact as a differential $(p+1)$ -form on $\text{vis}_x(M)$.

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